



**SCHOOL OF
COMPUTING
CHENNAI**

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #4

Activity Name: LCD Interfacing

Name of Student:	C S Deeraj
Registration No.:	CH.SC.U4CSE24106
Activity Date:	18/08/2026
Submission Date:	18/08/2026

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	10	

Faculty In-charge

Write an embedded C program for ARM architecture to interface LCD. Debug and verify the program using IDE on a development computer system.

Embedded C Code:

```
#include <LPC21xx.H>
#define RS 0x00000001 // P0.0->RS
#define RW 0x00000002 // P0.1->RW
#define EN 0x00000004 // P0.2->EN
void delay(unsigned int t)
{
    unsigned int i;
    for(i=0; i<t; i++);
}
void lcd_cmd(unsigned char cmd)
{
    IO1PIN = (IO1PIN & 0x00FFFFFF) | (cmd << 24); // Send command to P1.24-P1.31
    IO0CLR = RS; // RS = 0 for command
    IO0CLR = RW; // RW = 0 for write
    IO0SET = EN;
    delay(1000);
    IO0CLR = EN;
    delay(5000);
}
void lcd_data(unsigned char data)
{
    IO1PIN = (IO1PIN & 0x00FFFFFF) | (data << 24); //Send data to P1.24-P1.31
    IO0SET = RS; // RS =1 for data
    IO0CLR = RW;
    IO0SET = EN;
    delay(1000);
    IO0CLR = EN;
    delay(5000);
}
void lcd_init(void)
{
    lcd_cmd(0x38); //8 bit mode, 2 line, 5x7 font
    lcd_cmd(0x0C); //Display ON, cursor OFF
    lcd_cmd(0x01); //Clear display
    delay(5000);
    lcd_cmd(0x06); //Entry mode: increment cursor
    lcd_cmd(0x80); //cursor to first line
}
void lcd_string(char *str)
{

```

```

while(*str)
{
    lcd_data(*str);
    str++;
}
}
int main(void)
{
    PINSEL0 = 0x00000000;
    PINSEL2 = 0x00000000;
    IO0DIR = RS | RW | EN; //P0.0, P0.1, P0.2 as output
    IO1DIR = 0xFF000000; //P1.24 - P1.31 as output (data lines)
    lcd_init();
    lcd_cmd(0x80);          // Line 1, position 0
    lcd_string("Hey Guys It's ME");
    lcd_cmd(0xC0);          // Line 2, position 0 - 0xC0 is second-line address
    lcd_string("CSE24106 Deeru");

    while(1);
}

```

Evaluation mode output

C:\Users\CH.SC.U4CSE24106\Desktop\CH.SC.U4CSE24106 Embedded Systems\LCD lab ex 4\LCD exp.uvproj - µVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Current	
R0	0x40000060
R1	0x40000060
R2	0x40000060
R3	0x40000060
R4	0x00000000
R5	0x40000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00003370
R11	0x00000000
R12	0x000002E8
R13 (SP)	0x40000460
R14 (LR)	0x000000C0
R15 (PC)	0x000002E8
CPSR	0x600000D3
SPSR	0x00000000
User/System	
Fast Interrupt	
Interrupt	
Supervisor	
Abort	
Undefined	
Internal	
PC \$	0x000002E8
Mode	Supervisor
States	866
Sec	0.00007217

Disassembly

```

0x000002E4 00001388 ANDEQ R1,R0,R8,LSL #7
S3: IO1DIR = 0xFF000000; //P1.24 - P1.31 as output (data lines)
0x000002E8 E3A00000 MOV R0,#0x00000000
0x000002EC E59F1044 LDR R1,[PC,#0x0044]
0x000002F0 E5810000 STR R0,[R1]
S4: lcd_init();

```

lcd interface.cpp*

```

31 {
32     lcd_cmd(0x38); //8 bit mode, 2 line, 5x7 font
33     lcd_cmd(0x0C); //Display ON, cursor OFF
34     lcd_cmd(0x01); //Clear display
35     delay(5000);
36     lcd_cmd(0x06); //Entry mode: increment cursor
37     lcd_cmd(0x80); //cursor to first line
38 }
39 void lcd_string(char *str)
40 {
41     while(*str)
42     {
43         lcd_data(*str);
44         str++;
45     }
46 }
47 int main(void)
48 {
49     PINSEL0 = 0x00000000;
50     PINSEL2 = 0x00000000;
51     IO0DIR = RS | RW | EN; //P0.0, P0.1, P0.2 as output
52     IO1DIR = 0xFF000000; //P1.24 - P1.31 as output (data lines)
53     lcd_init();
54     lcd_cmd(0x80); // Line 1, position 0
55     lcd_string("Hey Guys It's ME");
56     lcd_cmd(0xC0); // Line 2, position 0 - 0xC0 is second-line address
57     lcd_string("CSE24106 Deeru");
58 }
59 while(1);
60 }

```

Project Registers

Other

Project Registers

Command

```

Running with Code Size Limit: 32K
Load "C:\Users\CH.SC.U4CSE24106\Desktop\CH.SC.U4CSE24106 Embedded Systems\LCD lab ex 4\Objects\LCD exp.axf"
*** Restricted Version with 32768 Byte Code Size Limit
*** Currently used: 880 Bytes (2%)

```

Call Stack - Locals

Name	Location/V...	Type
main	0x000002E8	int f0

ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE DEFINE DIR Display Enter EVALuate

Call Stack - Locals Memory 1

Real-Time Agent: Target Reset Simulation t1: 0.00007217

Output:

